

The Cambridge University HPC Service

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Bioinformatics Training

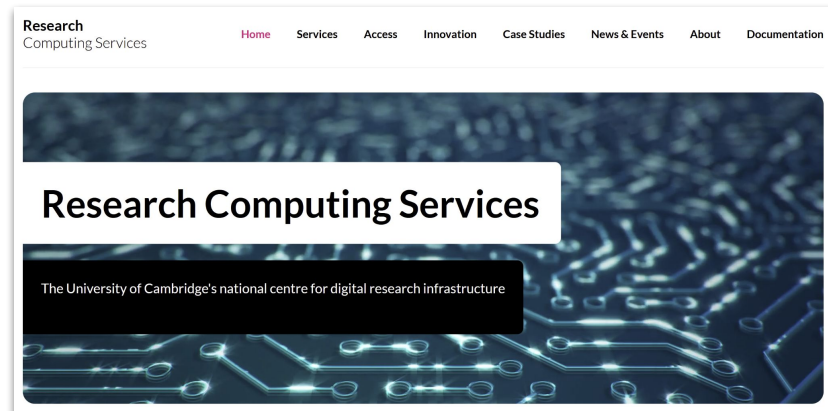
Types of nodes

Cambridge Service for Data-Driven Discovery (CSD3) consists of:

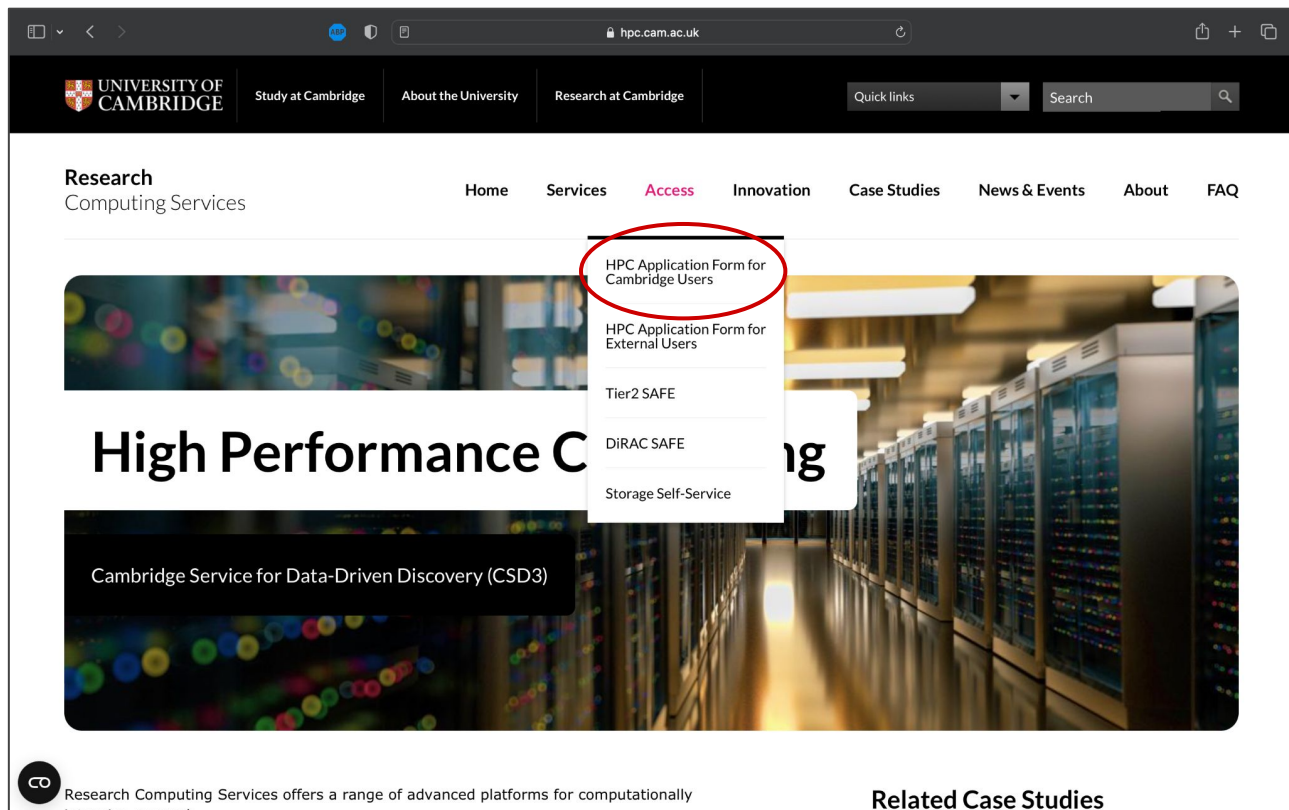
- CPU cluster → known as **Cumulus** with 3 partitions:
 - Cascade Lake – 56 CPUs, 192 GiB RAM (384 GiB for high-memory nodes)
 - Ice Lake – 76 CPUs, 256 GiB RAM (512 GiB for high-memory nodes)
 - Sapphire Rapids – 112 CPUs, 512 GiB RAM
- GPU cluster → known as **Wilkes3** with a single partition (`-p ampere`):
 - 4x NVIDIA A100-SXM-80GB GPUs

Documentation links

- Main site
<https://www.hpc.cam.ac.uk/>
- HPC docs:
<https://docs.hpc.cam.ac.uk/>
- Storage docs:
<https://docs.hpc.cam.ac.uk/storage/>



Register for an account



The screenshot shows the University of Cambridge Research Computing Services website. The browser address bar displays `hpc.cam.ac.uk`. The top navigation bar includes the University of Cambridge logo, links for 'Study at Cambridge', 'About the University', and 'Research at Cambridge', along with a 'Quick links' dropdown and a search bar. Below this, a secondary navigation bar features 'Research Computing Services' and a menu with 'Home', 'Services', 'Access' (highlighted in pink), 'Innovation', 'Case Studies', 'News & Events', 'About', and 'FAQ'. A dropdown menu is open under 'Access', listing 'HPC Application Form for Cambridge Users' (circled in red), 'HPC Application Form for External Users', 'Tier2 SAFE', 'DiRAC SAFE', and 'Storage Self-Service'. The main content area has a large banner with the text 'High Performance Computing' and a background image of server racks. Below the banner, a section titled 'Cambridge Service for Data-Driven Discovery (CSD3)' is visible. At the bottom, a footer contains the Research Computing Services logo and text, and a link to 'Related Case Studies'.

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HPC Application Form for Cambridge Users

HPC Application Form for External Users

Tier2 SAFE

DiRAC SAFE

Storage Self-Service

High Performance Computing

Cambridge Service for Data-Driven Discovery (CSD3)

Research Computing Services offers a range of advanced platforms for computationally

Related Case Studies

Register for an account

Your PI will receive an email to approve your registration (make sure you talk to them beforehand)

Internal HPC Application Form

This form should be completed by individual users with University login credentials (i.e. a Raven/UIS password) to apply for an account on the Research Computing Service systems. The Principal Investigator (PI) will be contacted via email to confirm the application.

The PI of the project should have the authority in principle to release grant funds; research students are recommended to approach their supervisors for advice regarding the appropriate person in their research group. Note that a project may already exist to cover similar work.

Before entering personal data into this form, please review the [Research Computing Services Privacy Policy](#). You will need to consent to this policy before submitting your data.

Principal Investigator Information

PI Name *

PI CRSid *

The PI's CRSid (unique University of Cambridge identifier, e.g. abc123).

PI Status *

e.g. Professor, Lecturer, ...

PI Email

Register for an account

Service Level *

Note it is only possible to add users to pre-existing paying projects through this form. If you are a PI wishing to purchase paying resources please contact us separately via support@hpc.cam.ac.uk.

- ☐ Non-paying (SL3) only
- ☐ Non-paying (SL3) plus paying (SL2)
- ☐ Non-paying (SL3) plus paying (SL1)

End Date

dd / mm / yyyy



Compute Platforms

By default new users will receive computational resources on the Peta4-Skylake CPU cluster. For more information please see <https://www.hpc.cam.ac.uk/systems/peta-4>.

- ☐ Please indicate below which specialist CSD3 platforms are also required (if any)
- Wilkes3-GPU

Please indicate to which dedicated nodes (if any) you have been advised you should have access *

- ☐ None
- ☐ Materials Theory Group (Professor Chris Pickard)
- ☐ MRC Epidemiology Unit
- ☐ EpiCov CUH

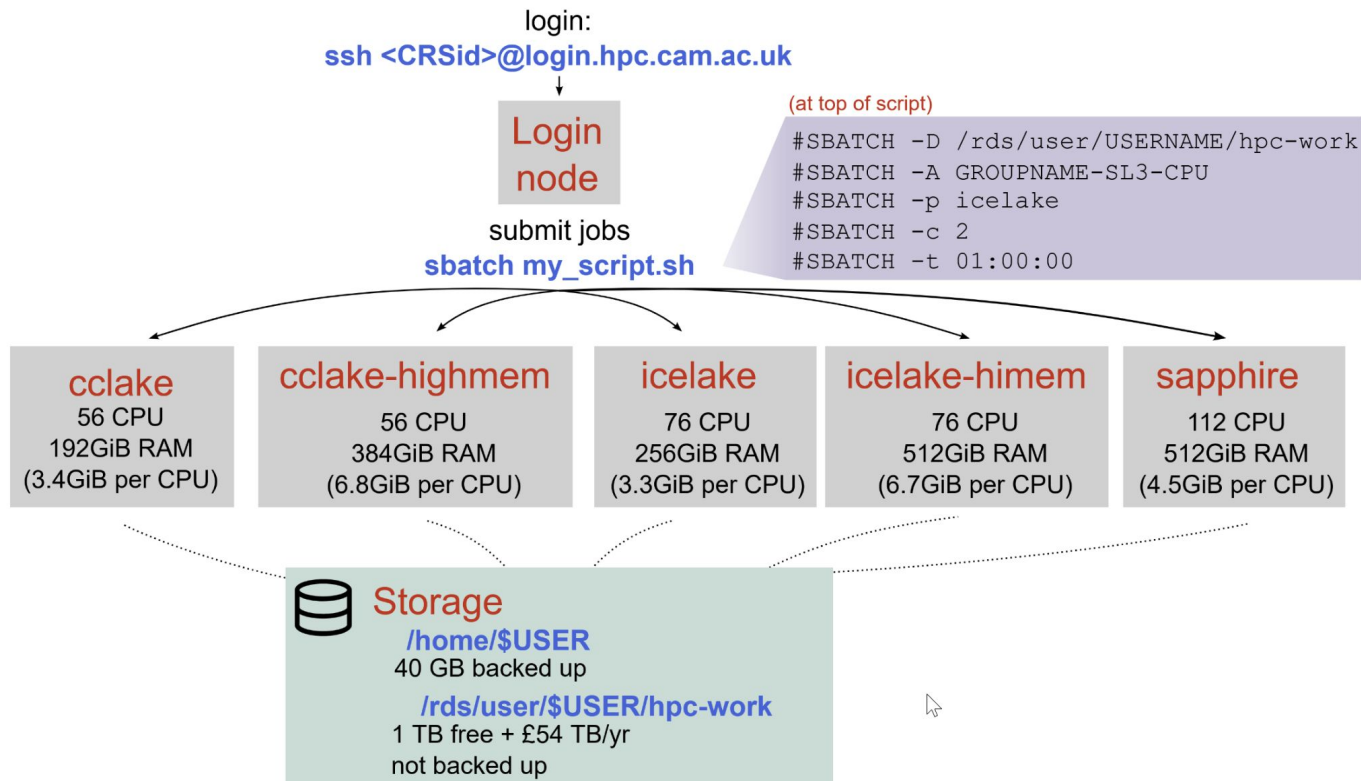
Service levels

	SL3	SL2
Cost	Free: each PI has 200,000 CPU hours and 3000 GPU hours per quarter	Paid per unit hour (CPU: £0.01; GPU: £0.550)
Priority	Lower priority	Higher priority
Time limit	12h	36h

You can check your accounts and credits with **mybalance** command

More about policies [in the docs](#).

Overview of CPU cluster



Charging model - SL2 (paid service)

- **CPU nodes** → currency is **CPU hour** (£0.01)
 - Job runs for 2h with 1 CPUs = 2 CPU hours = £0.02
 - Job runs for 2h with 5 CPUs = 10 CPU hours = £0.10
 - Job runs for 36h with 76 CPUs = 2736 CPU hours = £27.36
- **GPU nodes** → currency is **GPU hour** (£0.55)
 - Job runs for 2h with 1 GPU = 2 GPU hours = £1.1
 - Job runs for 36h with 4 GPUs = 144 GPU hours = £79.2

CPU partitions (`icelake`, `cclake`, `sapphire`)

Memory is tied to CPU

- E.g. on `icelake` we have ~3.3G per CPU
 - User requests `-c 3` they will get ~9.9G memory
 - User requests `--mem 15G` they will get 5 CPUs
- Therefore for memory-intensive tasks it's more cost effective to use the `himem` partitions

GPU partitions (ampere)

- Each node has 4 GPUs (you get 32 CPUs per GPU)
- Resource request:

```
#SBATCH -A PROJECT-SL3-GPU  
#SBATCH -p ampere  
#SBATCH --nodes=1  
#SBATCH --gres=gpu:3
```

→ You need to specifically have a GPU account active (check with **mybalance**; if not, request from support@hpc.cam.ac.uk)

→ This would request 3 (of the 4 max) GPUs in the node.

Cambridge HPC filesystem

- Home directory has very limited space (50GB)
- `~/rds/hpc-work/` is the main working folder with 1TB storage limit (not backed up)
- `~/rds/hpc-work/` folder is not shareable
 - If you work in a team, the PI needs to buy additional RDS (research data storage) space → <https://selfservice.uis.cam.ac.uk/>
- Because of MFA *Filezilla* needs to be configured [as detailed here](#)

Module system for applications

- List all available modules

```
module avail
```

```
module avail -a -i "appname"
```

- Load in modules and test them on the login node

Example: `module load gromacs/2021/cclake`

- Remember to load modules in your sbatch script
- Use modules instead of installing them to your own storage space
- You can ask HPC support to install new modules

Additional Advice

- Create some universal sbatch request sets
 - E.g. cclake-1h-4cpu
- Use modules instead of downloading / installing software
 - "module avail" command for checking availability
- Pass argument to the sbatch request set
 - SampleID, input files
- Redirect standard output and error to specific files
 - #SBATCH -o and #SBATCH -e
- Build dependency chains and do checkpointing if the 12/36h time limit is not enough
- Interactive jobs → documentation!

Additional Advice

- Check your storage status
 - `quota`
- Check your account information (how many CPU hours you have)
 - `mybalance`
- Environments / Containers
 - No Docker allowed, you can use Singularity instead
 - Conda/Mamba is go solution for local package management
- Don't abuse login nodes → you may approach others if they are doing it
- Report problems to the `support@hpc.cam.ac.uk`

Use login nodes ethically

```
Tasks: 1332 total, 27 running, 1302 sleeping, 3 stopped, 0 zombie
%Cpu(s): 47.1 us, 12.0 sy, 0.0 ni, 1.6 id, 39.2 wa, 0.0 hi, 0.0 si, 0.0 st
KiB Mem : 19647073+total, 36855588 free, 80972224 used, 78642920 buff/cache
KiB Swap: 15999996 total, 11342332 free, 4657664 used. 11469084+avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
192127	dc124	20	0	845952	212836	11620	R	99.4	0.1	0:04.04	numerical_gr
152402	rm889	20	0	4772	672	468	R	94.5	0.0	46:15.15	gzip
84900	rm889	20	0	4772	676	468	D	46.0	0.0	217:49.73	gzip
84902	rm889	20	0	4772	676	468	D	46.0	0.0	217:46.97	gzip
10212	rm889	20	0	4772	676	468	R	45.4	0.0	217:49.28	gzip
10213	rm889	20	0	4772	676	468	R	45.4	0.0	217:48.47	gzip
84899	rm889	20	0	4772	672	468	D	45.4	0.0	217:39.52	gzip
84901	rm889	20	0	4772	676	468	S	45.4	0.0	218:04.45	gzip
129135	rm889	20	0	4772	676	468	S	45.4	0.0	218:00.44	gzip
129136	rm889	20	0	4772	676	468	R	45.4	0.0	217:59.99	gzip
10211	rm889	20	0	4772	676	468	R	44.8	0.0	217:48.16	gzip
10218	rm889	20	0	4772	676	468	D	44.8	0.0	264:59.32	gzip
10219	rm889	20	0	4772	676	468	R	44.8	0.0	265:12.54	gzip
10215	rm889	20	0	4772	676	468	S	44.2	0.0	264:55.00	gzip
10216	rm889	20	0	4772	676	468	D	44.2	0.0	265:05.74	gzip
10217	rm889	20	0	4772	676	468	R	44.2	0.0	264:53.97	gzip
10220	rm889	20	0	4772	676	468	S	44.2	0.0	264:44.60	gzip
84908	rm889	20	0	4772	676	468	R	44.2	0.0	264:55.98	gzip
84907	rm889	20	0	4772	672	468	D	43.6	0.0	265:03.41	gzip
84909	rm889	20	0	4772	676	468	D	43.6	0.0	265:11.01	gzip
211690	rm889	20	0	4772	676	468	D	20.2	0.0	58:41.63	gzip
211682	rm889	20	0	4772	676	468	D	19.6	0.0	58:41.09	gzip
211683	rm889	20	0	4772	676	468	D	19.0	0.0	58:42.18	gzip
211684	rm889	20	0	4772	676	468	D	19.0	0.0	58:42.06	gzip
211685	rm889	20	0	4772	676	468	D	19.0	0.0	58:41.40	gzip
211686	rm889	20	0	4772	676	468	D	19.0	0.0	58:42.32	gzip
211691	rm889	20	0	4772	676	468	D	19.0	0.0	58:39.96	gzip
211687	rm889	20	0	4772	676	468	D	18.4	0.0	58:40.93	gzip
211688	rm889	20	0	4772	676	468	D	18.4	0.0	58:41.91	gzip
211689	rm889	20	0	4772	672	468	D	18.4	0.0	58:40.53	gzip
1048	rm889	20	0	4772	676	468	R	16.6	0.0	89:00.86	gzip
1051	rm889	20	0	4772	672	468	D	16.6	0.0	88:52.42	gzip
114091	rm889	20	0	4772	676	468	D	16.6	0.0	88:48.15	gzip
1049	rm889	20	0	4772	672	468	R	16.0	0.0	88:54.63	gzip
1050	rm889	20	0	4772	676	468	D	16.0	0.0	88:46.55	gzip
114092	rm889	20	0	4772	676	468	D	16.0	0.0	88:57.99	gzip
157027	rm889	20	0	4772	676	468	R	16.0	0.0	89:00.51	gzip
157029	rm889	20	0	4772	676	468	R	16.0	0.0	88:47.60	gzip
1053	rm889	20	0	4772	676	468	D	15.3	0.0	88:58.61	gzip
157028	rm889	20	0	4772	676	468	R	15.3	0.0	88:50.95	gzip
255619	rm889	20	0	4772	676	468	D	11.0	0.0	90:12.48	gzip

Email Aug 2022 from HPC Service Desk:

Please can I remind everyone that the login nodes are not intended as a means to avoid the queueing system and that all compute intensive applications should be submitted to Slurm. Only small and brief tests may be run directly on the login nodes.

A few minutes ago, login-q-4 was brought briefly to its knees by multiple users running matlab, node, and python processes leading to all cpus and all memory being engaged. This was an inconvenience to other users on the node, and also led to the destabilisation of the login node, requiring manual intervention. If you wish to use a large proportion of resources on a node, please request a compute node through Slurm.

Please see our documentation on interactive use for more information:

<https://docs.hpc.cam.ac.uk/hpc/user-guide/interactive.html>

Thankyou for your attention.

Do not use `/home` for intensive jobs

Email Apr 2023 from HPC Service Desk:

The problems which began around midday [frozen home directories to many users] appear to have been caused by a user submitting I/O intensive python jobs reading from /home on multiple cpus on around 100 nodes.

Please can I remind everyone that /home is not designed for intense concurrent I/O from many nodes, and that reading python code is still I/O.

Also the consequences of overloading the NFS server behind /home and /usr/local are global and amount to a denial of service.

Currently nodes which are still alive are responding normally again, however there are a great many nodes (including some login nodes) which have not yet recovered. These will need to be rebooted.

Until the residual problem nodes have been cleared out, Slurm will not start new jobs.

Don't use unprotected credentials

--- bioinf@tup211:/home --- bash									
top - 16:32:27 up 46 days, 5:59, 99 users, load average: 4.13, 3.87, 3.10									
Tasks: 2321 total, 4 running, 2311 sleeping, 4 stopped, 2 zombie									
MiB Mem : 2.9 us, 3.9 sy, 0.0 ni, 89.4 id, 2.6 wa, 0.0 hi, 0.2 si, 0.0 st									
KiB Mem : 1946476 total, 746168 free, 4396726 used, 277848 buff/cache									
KiB Swap : 19599996 total, 2738188 free, 13261808 used, 15861318 avail Mem									
PID USER PR NI VIRT RES SHR S CPU MEM TIME COMMAND									
95288	jso57	20	0	137448	2876	1188	26.0	0.0	19:47.40 rsync --server --vlog@prz /zds/project/fm66/rds-fm66-seq-learning/30_backups/jo_backziped18_01_22.zip
73111	efn30	20	0	153188	2749	2892	24.0	0.0	33:57.64 wget -c -user emckinney -password http://sunstrider.medgen.medschl.cam.ac.uk/emckinney/220118_A01303_0054_B00HVTVDSX2_emckinney.tar.gz
145889	sjso80	20	0	157137	56474	18336	1.0	0.1	15:18.37 python3.6 -w /home/cis-torati/i-pi/bin/i-pi INITIAL.xml
68789	efn30	20	0	153188	2736	2892	6.7	0.0	34:33.05 wget -c -user emckinney -password http://sunstrider.medgen.medschl.cam.ac.uk/emckinney/220118_A01303_0055_AHW33DSX2_emckinney.tar.gz
235724	sjso80	20	0	193668	8428	3276	6.6	0.0	3:32.97 ssh -X cpu-p460
138642	jc2113	20	0	293720	21696	6416	0.1	0.0	12:10.92 rstudio
280125	lk397	20	0	176424	8444	2148	8.0	0.0	0:02.28 top
93641	root	20	0	0	0	0	2.6	0.0	0:15.63 [worker/u516:3]
95287	jso57	20	0	194672	4516	1148	2.6	0.0	1:42.82 sshd: jso57@otty
126809	sjso80	20	0	192884	2768	1120	5.0	0.0	1:01.74 sshd: sjso80@otty
201434	root	20	0	49688	1556	1188	1.6	0.0	0:00.05 /bin/ps -p 4790 -o args=
111892	cis-torati	20	1	1148436	192374	18336	0.1	0.1	14:48.37 python3.6 -w /home/cis-torati/i-pi/bin/i-pi INITIAL.xml
265449	rm617	20	0	991536	181820	17312	1.0	0.1	25:05.00 kdeinit4: kwallet [kdeinit]
9	root	20	0	0	0	0	0.6	0.0	0:22:33.77 [rcu_sched]
3852	root	20	0	0	0	0	0.6	0.0	0:22:35.92 [sbin/rund -f
20779	root	0	-20	0	0	0	0.6	0.0	0:01.04 [worker/20:0H]
46934	root	0	-20	0	0	0	0.6	0.0	0:32.29 [worker/2:0H]
118353	sjso54	20	0	972788	142732	20880	0.0	0.1	11:44.89 /home/sjso54/.vscode-server/bin/899d46d82c4c95423fb7e10e68ba52050e38ba3/node /home/sjso54/.vscode-server/bin/899d46d82c4c95423fb7e10e68ba52050e38ba3/out/bootstrap-fork --type=extensionHost --urlTransf=
121151	rm617	20	0	502808	13620	4828	5.0	0.0	6:28.82 wish0.5 /Applications/Analyze/V12/Analyze-12.0/bin/AnalyzeMain.tbc
138496	root	20	0	0	0	0	0.6	0.0	0:58.27 [worker/4:2]
109861	root	20	0	0	0	0	0.6	0.0	0:52.87 [worker/u516:1]
238492	sjso80	20	0	3415264	420188	166716	5.0	0.2	2:37.34 /usr/local/software/fsl/6.0.4/fslpython/envs/fslpython/bin/python /usr/local/software/fsl/6.0.4/bin/fslmeyes
2	root	20	0	0	0	0	0.3	0.0	0:42.76 [kthread]
59	root	20	0	0	0	0	0.3	0.0	18:38.78 [ksftirqd/2]
26	root	20	0	0	0	0	0.3	0.0	9:38.79 [ksftirqd/3]
86	root	20	0	0	0	0	0.3	0.0	0:54.89 [ksftirqd/15]
2551	root	-51	0	0	0	0	0.3	0.0	22:58.33 [irc/655-115.0 1
4942	root	20	0	1227636	5812	3252	5.0	0.0	15:00.31 /usr/sbin/automount --systemd-service --dont-check-daemon
12602	root	20	0	0	0	0	0.3	0.0	0:12:23.70 [ptlpd_00_15]
20287	dc-cad11	20	0	42880	28992	996	0.3	0.0	7:14.41 tmux
31836	sjso80	20	0	194652	8484	3272	5.0	0.0	1:19.61 ssh -X cpu-p460
49943	rm294	20	0	27248	560	372	0.3	0.0	18:07.31 /usr/libexec/gdm_server
49917	root	20	0	0	0	0	0.7	0.0	1:16.31 [idle 31]
114829	wjml4	20	0	1001916	42820	17492	5.0	0.0	0:34.86 /home/wjml4/.vscode-server/bin/899d46d82c4c95423fb7e10e68ba52050e38ba3/node /home/wjml4/.vscode-server/bin/899d46d82c4c95423fb7e10e68ba52050e38ba3/out/bootstrap-fork --type=ptyHost
146574	sjso54	20	0	239224	40760	7440	5.0	0.0	7:34.08 /usr/lib64/bin/sxsc/R --silent --save --no-restore -f /home/sjso54/.vscode-server/extensions/kyuadeu-2.3.6/R/help/HelpServer.R
176728	j7699	20	0	888020	25932	5360	1.0	0.1	0:19:56 /usr/bin/vscode-server/bin/899d46d82c4c95423fb7e10e68ba52050e38ba3/node /home/j7699/.vscode-server/bin/899d46d82c4c95423fb7e10e68ba52050e38ba3/out/vs/server/main.js --start-server --host=127.0.0.1
138743	jc2113	20	0	1452616	808452	23824	0.3	0.4	18:43.36 /usr/local/Cluster-Apps/rstudio/1.1.383/bin/session --config-file none --program-mode desktop --www-port 20591 --launcher-token B208DA11 --show-help-home 1
265137	jeff86	20	0	538936	19732	12852	0.3	0.0	0:00.39 xfdesktop
81168	root	0	-20	0	0	0	0.3	0.0	0:22.22 [worker/4:0H]
193759	root	20	0	0	0	0	0.3	0.0	0:01.33 [worker/6:2]
194826	v1571	20	0	128724	4780	1860	0.3	0.0	0:00.23 bash
196616	root	20	0	133420	1624	1312	5.0	0.0	0:00.57 /bin/bash /usr/local/Cluster-Apps/watchdog/bin/watchdog.sh
200296	root	20	0	0	0	0	0.3	0.0	0:00.02 [worker/8:2]
217280	lc800	20	0	113424	1644	1276	0.6	0.0	0:01.06 /bin/bash /usr/bin/s2psresume-session lc800-60-1642440216_stdXFC_dp32 1024x768 adsl 16m-jpeg-9 us query 0 both no
255183	rm617	20	0	862392	28220	19784	5.0	0.0	1:49.45 kdeinit4: kwallet [kdeinit]
1	root	20	0	191968	3852	1728	5.0	0.0	15:35.05 /usr/lib/systemd/systemd --switched-root --system --deserialize 22
4	root	0	-20	0	0	0	0.6	0.0	0:00.00 [worker/0:0H]
6	root	20	0	0	0	0	0.6	0.0	1:45.82 [ksftirqd/0]
7	root	rt	0	0	0	0	0.6	0.0	0:00.61 [migration/0]
8	root	20	0	0	0	0	0.6	0.0	0:00.00 [rcu_bh]
10	root	0	-20	0	0	0	0.6	0.0	0:00.00 [lru-add-drain]
11	root	rt	0	0	0	0	0.6	0.0	0:13.52 [watchdog/0]
12	root	rt	0	0	0	0	0.6	0.0	0:15.74 [watchdog/1]
13	root	rt	0	0	0	0	0.6	0.0	0:44.23 [migration/1]
14	root	20	0	0	0	0	0.6	0.0	0:49.57 [ksftirqd/1]
16	root	0	-20	0	0	0	0.6	0.0	0:00.00 [worker/2:0H]
18	root	rt	0	0	0	0	0.6	0.0	0:12.53 [watchdog/2]
19	root	rt	0	0	0	0	0.6	0.0	1:00.25 [migration/2]
24	root	rt	0	0	0	0	0.6	0.0	0:28.99 [watchdog/3]
25	root	rt	0	0	0	0	0.6	0.0	1:44.59 [migration/3]
28	root	0	-20	0	0	0	0.6	0.0	0:00.00 [worker/3:0H]
29	root	rt	0	0	0	0	0.6	0.0	0:12.28 [watchdog/4]
30	root	rt	0	0	0	0	0.6	0.0	0:25.10 [migration/4]
31	root	20	0	0	0	0	0.6	0.0	3:54.21 [ksftirqd/4]
34	root	rt	0	0	0	0	0.6	0.0	0:16.44 [watchdog/5]
35	root	rt	0	0	0	0	0.6	0.0	0:55.66 [migration/5]
36	root	20	0	0	0	0	3.01	11	[ksftirqd/5]

Don't use unprotected credentials

```
%MEM      TIME+  COMMAND
0.0 19:47.98 rsync --server --sender -vlogDtpzr . /rds/project/fm456/rds-fm456-seq-learning/JO_backups/jo_backzipped18_01
0.0 33:57.64 wget -c --user emckinney --password [REDACTED] http://sunstrider.medgen.medschl.cam.ac.uk/emckinney/2201
0.3 157:18.45 x2goagent -nolisten tcp -nolisten tcp -dpi 96 -D -auth /home/sjs80/.Xauthority -geometry 1920x1040 -name X2G
0.0 34:33.05 wget -c --user emckinney --password [REDACTED] http://sunstrider.medgen.medschl.cam.ac.uk/emckinney/2201
0.0 3:32.97 ssh -X cpu-p-460
0.1 12:10.02 rstudio
0.0 0:02.28 top
0.0 0:15.63 [kworker/u514:3]
0.0 1:42.02 sshd: iag57@patty
```

Tips

- For quick testing you can get highest-priority (even on SL3) by adding:

```
#SBATCH --qos=intr
```

However:

- These jobs are limited to 1h
 - You can only have 1 of them at any given time
- Create aliases on your `~/ .bashrc` for commonly used commands ([see here](#) if you don't know what they are).

For example:

```
alias nodestat='sinfo -o "%15P %.5a %.6D %.6t" '  
alias myqueue='squeue -u $USER'
```

Use the HPC...

- ... **Ethically**

- Do not use the login node for production runs

- ... **Smartly**

- Optimise your jobs for CPU and time usage
- Create universal and software specific submission scripts (but never sample specific)
- Reduce the number of CPU cores if it doesn't have a very significant effect to go in production earlier
- Check production node usage (`sinfo -o "%15P %.5a %.6D %.6t"`)

- ... **Efficiently**

- Use SL3 if the job is not urgent
- Run multiple samples in parallel
- Build up dependency chains